

Trainings ▾

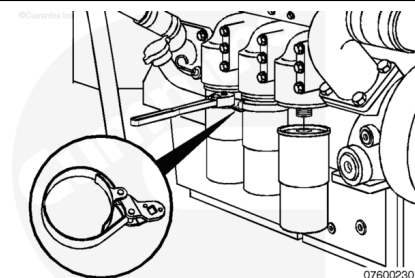
Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



The QSK45 engine has three combination oil filters and the QSK60 has four combination oil filters.

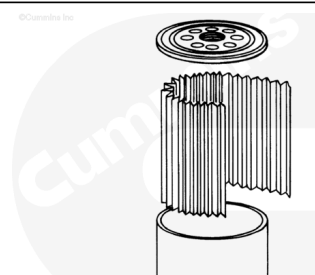
Use an oil filter wrench, Part Number 3375049, or equivalent to remove the oil filters.

Discard the filter if **not** required for failure analysis.

Inspect

⚠ WARNING ⚠

Carefully cut the combination oil filter open. The filter element spring is under compression and can cause personal injury.



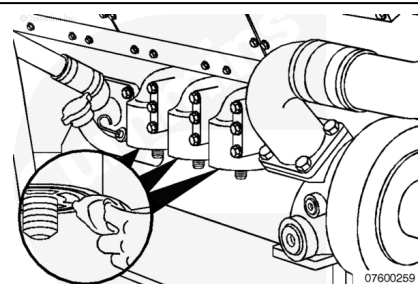
Use tube cutter, Part Number 3376579, to open the full-flow oil filter.

Inspect the filter element for evidence of moisture and metal particles.

Metal	Probable Source
Copper	Bearings and bushings
Chromium	Piston rings
Iron	Cylinder liners, gears, and camshafts
Lead	Bearing overlay material
Aluminum	Piston skirts, camshaft bushings, crankshaft thrust bearings, and aluminum aftercoolers

Install

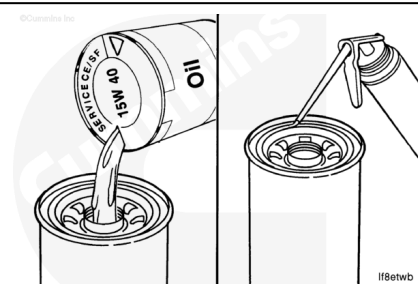
Clean the oil filter head sealing surface with a lint free cloth.



The QSK45 engine has three combination oil filters and the QSK60 engine has four combination oil filters.

Use engine oil to lubricate the oil filter's rubber seal.

Fill the oil filters with clean engine oil.

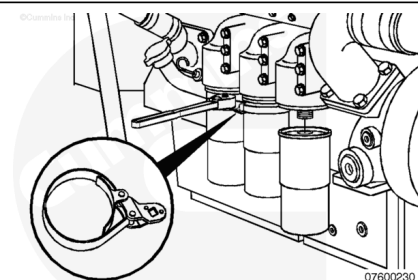


⚠ CAUTION ⚠

Mechanical overtightening can distort the threads or damage the filter element seal. If the threads appear damaged, replace the oil filter head adapter.

⚠ CAUTION ⚠

Locate the oil filter wrench, Part Number 3375049, near the top of the oil filter canister. This will reduce the possibility of damaging the oil filter.

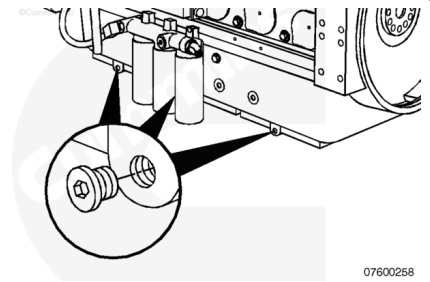


Install the oil filter turning it until the seal contacts the filter head's sealing surface.

Turn the oil filter an additional 3/4 to 1 turn.

Install and tighten the drain plug.

Torque Value: 47 N·m [35 ft-lb]



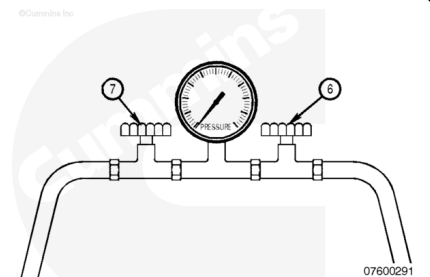
Pressure Differential Test

Spin-on Type

This test will indicate oil pressure before and after the oil flows through the filters.

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves to eliminate gauge error. Use a gauge with a minimum pressure capacity of 1400 kPa [200 psi].

Be sure the engine is switched off.

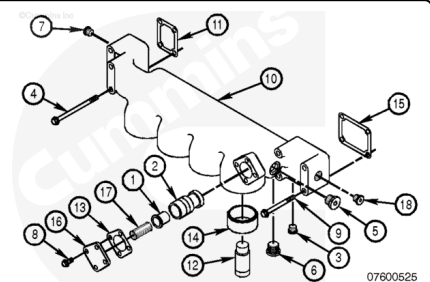


Remove the M14 hex plug (7) from the outlet end of the oil filter head (10) and install end 7 of the valve assembly in the port of the filter head from where the hex plug (7) has been removed.

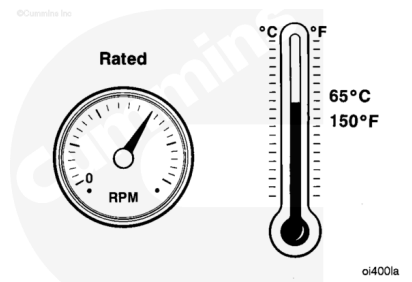
This reading will indicate the oil pressure after the oil flows through the filters.

Remove the hex plug (18) from the oil pump inlet end of the filter head (10) and install end 6 of the valve assembly oil hose into where the second hex plug (18) has been removed.

This reading will indicate the oil pressure before the oil flows through the filters.



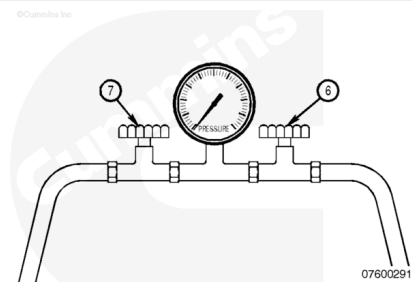
Operate the engine at rated rpm until the engine oil reaches its normal operating temperature. Continue running the engine at rated rpm for the duration of the pressure test.



Close valve (7) and open valve (6) to read the oil pressure before the filters.

Close valve (6) and open valve (7) to read the oil pressure after the filters.

If the difference in pressure is more than the maximum allowable value, an oil filter with excessive restriction is being used. Replace the filters and check the pressure differential again.



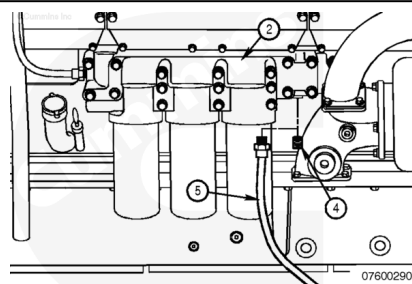
Measurements			
		kpa	psi
Maximum Oil Filter Pressure Differential	New/clean oil filters	83	12
	Used oil filters	240	35

Switch off the engine.

Remove the end of the valve assembly (5) from the oil pump end of filter head (2) and install the hex plug (4).

Tighten the hex plug.

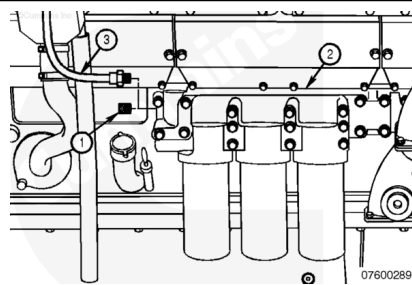
Torque Value: 27 n•m [20 ft-lb]



Remove the hose (3) from outlet end of filter head (2) and install the hex plug (1).

Tighten the hex plug.

Torque Value: 27 n•m [20 ft-lb]

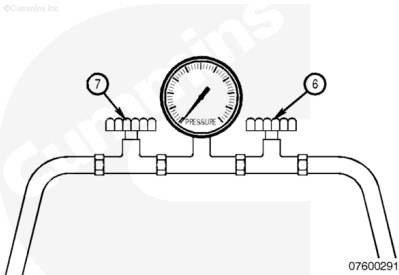


Eliminator

This test will indicate oil pressure before and after the oil flows through the eliminator.

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves to eliminate gauge error. Use a gauge with a minimum pressure capacity of 1400 kPa [200 psi].

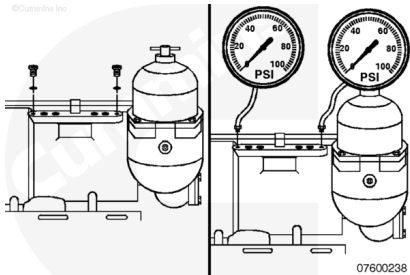
Be sure the engine is switched off.



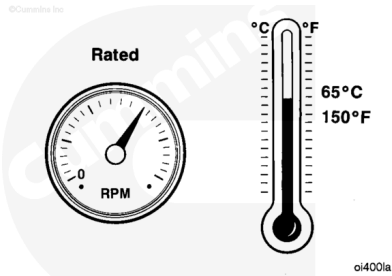
Remove the two plugs from the filter housing.

Install end 6 of the valve assembly in the port of the filter nearest to the centrifuge from where the plug has been removed.

Install end 7 of the valve assembly in the port of the filter furthest away from the centrifuge from where the other plug has been removed.



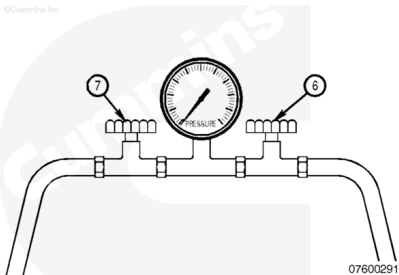
Operate the engine at rated rpm until the engine oil reaches its normal operating temperature. Continue running the engine at rated rpm for the duration of the pressure test.



Close valve (7), open valve (6), and record the pressure value before the eliminator.

Close valve (6), open valve (7), and record the pressure value after the eliminator.

If the difference in pressure is more than the maximum allowable value, inspect and clean the eliminator screens before checking the oil pressure differential again.



Measurements

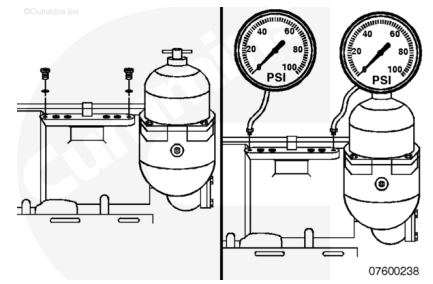
	kpa	psi
Maximum Oil Pressure Differential	60	9

Switch off the engine.

Remove the valve assembly from the eliminator and reinstall the two plugs.

Tighten the plugs.

Torque Value: 10 n•m [84 in-lb]



Last Modified: 14-Sep-2007
